

### The Utility Problem

A transformer load refers to the amount of power that is being drawn by the electrical load connected to a transformer. It is important to ensure that a transformer is not overloaded, as this can cause damage to the transformer and potentially lead to power outages or other issues. Overheating and, consequently, reduced transformer lifespan are some of the consequences of an overloaded transformer.

With an increasing number of requests for new connections in the distribution grid, such as businesses, households, EV chargers etc., it is important to have a capacity management plan in place to ensure that the distribution grid can accommodate the new connections without compromising the reliability and quality of service for existing customers.

It is, therefore, necessary to assess the load changes with these new loads included. If we use only the reserved capacities of the customers in the calculation, this calculation will not take into account the reality, but will only be an estimate, which is also likely to be very inaccurate, because the calculation will not include real measured data. As often there is no measurement at the transformer, and only individual meters "behind" the transformer are measured, one has no idea of the transformer load. However, information about the transformer load is very important, as replacing the transformer earlier with a new transformer with a higher capacity would require a large investment cost, which would be unnecessary if the transformer capacity is sufficient.

### The Data Solutions and Methods

This use case will greatly speed up the work of an analyst who would have to search for all the data from different data sources and connect and correct them, but in the Awesense Energy Transition Platform, the analyst has all the cleaned and synchronized data in one place and can work with it immediately.

For conducting such an analysis, it is necessary to have several datasets ready. These datasets are as follows:

1. Transformer asset information (properties of the transformer), specifically the transformer capacity - kVA rating.
2. Knowledge of the connected consumption points (loads) to the transformer - this information can be obtained very easily using the trace function - `grid_get_downstream()`.
3. Knowledge about the current load:
  - a. at the transformer (load time-series data) - if the measurement is at the transformer
  - b. or of the connected consumption points (load time-series data) - for each consumption point according to point 2, consumption data can be obtained using the function `ts_data_source_select()`.

## Transformer Load Increase Assessment Tool

### [UC38-01] - Awesense Energy Transition Platform Use Case

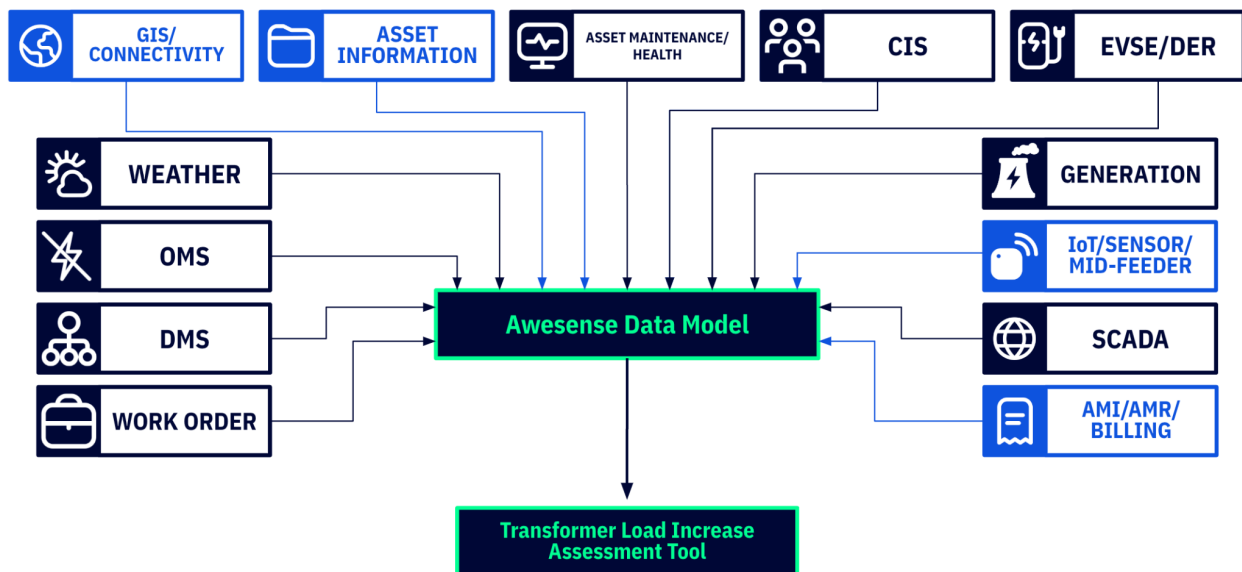


4. Knowledge about the future load - typical load profiles of consumption points (time-series data) - this information can be obtained from the use case "Standard Consumption Patterns Creation".

The calculation itself is, therefore, a comparison of the sum of all the consumption points "behind" the transformer plus future (added) consumption. These values are compared to the maximum capacity of the transformer for each time stamp. If the capacity is exceeded then a new load can't be connected to the given transformer.

### Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



### Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. A GitHub link to the Jupyter notebook can be provided upon request.